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Reference Values and Association of Body Weight, Age, and Sex With Echocardiographic Measurements in Non-Athletic Quarter Horses

Madeline E. Selecky[1] | Elizabeth Williams Louie[2] | Callum Donnelly[2] | Carrie J. Finno[3] | Jessica M. Morgan[1]

[1]Department of Medicine and Epidemiology, School of Veterinary Medicine, University of California-Davis, Davis, California, USA | [2]Department of Clinical Sciences, College of Veterinary Medicine, Cornell University, Ithaca, New York, USA | [3]Department of Population Health and Reproduction, School of Veterinary Medicine, University of California-Davis, Davis, California, USA

Correspondence: Jessica M. Morgan (jmmorgan@ucdavis.edu)

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ABSTRACT

Background: Echocardiographic reference intervals for Quarter Horses are infrequently reported.

Objectives: To provide standard echocardiographic measurements for sedentary Quarter Horses and evaluate the relationship between physical characteristics (body weight, age, sex) and echocardiographic measurements.

Animals: Forty-one sedentary Quarter Horses, free of cardiac disease, from a university research herd.

Methods: A cross-sectional study evaluating standardized echocardiograms performed on sedentary Quarter Horses (41) between 2020 and 2022. Reference intervals were calculated for raw and weight-corrected values. Multiple linear regression was performed to evaluate the relationship between the physical characteristics (body weight, age, sex) and echocardiographic variables.

Results: Multiple linear regression showed that every 100 kg of body weight was associated with a 3 (95% CI 1–6) mm thicker left ventricular free wall in systole ($p = 0.02$), an 8 (95% CI 2–13) mm larger left atrial diameter ($p < 0.01$), and a 4 (95% CI 0–9) mm larger pulmonary artery diameter ($p = 0.04$). Every year in age was associated with a 0 (95% CI 0–1) mm larger left atrial diameter ($p = 0.02$), a 0 (95% CI 0–1) mm larger aortic diameter ($p = 0.01$), and a 0 (95% CI 0–1) mm larger pulmonary artery diameter ($p = 0.01$). Left atrial to left ventricular ratio was 1 (95% CI −1 to 0) mm smaller in geldings compared to mares ($p = 0.03$).

Conclusions and Clinical Importance: This study provides echocardiographic reference intervals for a cohort of sedentary Quarter Horses and clarifies the association with body weight, age, and sex with these variables.

Abbreviations: 2D, two-dimensional; 4C, four-chamber; ANOVA, analysis of variance; AoD, aortic diameter; BWT, body weight; d, diastolic; FS, fractional shortening; G, gelding; IVS, interventricular septum thickness; L, left parasternal; LAAD, left atrial appendage diameter; LAD, left atrial diameter; L-AO, left parasternal long axis aortic; L-LA, left parasternal long axis left atrial; L-PA, left parasternal long axis pulmonic valve; LVFW, left ventricular free wall thickness; LVID, left ventricular internal diameter; LVOT, left ventricular outflow tract; M-mode, motion mode; MWT, mean wall thickness; *n*, number of horses; PAD, pulmonary artery diameter; R, right parasternal; R^2, coefficient of determination of the statistical model; RVOT, right ventricular outflow tract; RWT, relative wall thickness; s, systolic; SAX-AO, right parasternal short-axis view of the left ventricle at the level of the chordae tendineae; SAX-LV, right parasternal short-axis view of the left ventricle; SAX-MV, right parasternal short-axis view of the mitral valve; SD, standard deviation.

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1 | Introduction

Echocardiography is widely used to characterize and track the progression of cardiac disease in the horse [1]. Assessment of great vessel and chamber size is a critical tool to diagnose, stage, and monitor cardiac disease. Normal echocardiographic dimensions vary between breeds, and it is important for clinicians to be familiar with the normal range for each breed to accurately interpret echocardiographic findings [1, 2]. Quarter Horses' well-developed triceps musculature and conformation of the shoulder can make echocardiography more challenging than in some of the more well-described breeds, such as Thoroughbreds and Warmbloods. Friesians have smaller left ventricular measurements than expected for body size compared to Warmbloods, suggesting breed might be an important factor in echocardiographic dimensions [3]. There is a strong, positive correlation between body size and cardiac dimensional measurements in horses [2]. Although sex and age are associated with heart size in some species, such as humans and rats, this has not been consistently shown in horses [2, 4, 5].

Cardiac disease is an increasingly recognized cause of poor performance and systemic disease in the horse [6]. Risk assessment for both horse and rider/handler is often challenging, but imperative. Most current equine cardiac scientific research has been performed in athletic Thoroughbred, Standardbred, and Warmblood horses [3, 7–10]. Echocardiographic measurements have been infrequently reported for Quarter Horses [11–16]. Establishing normal cardiac dimensions in sedentary and recreational riding Quarter Horses will improve the assessment of cardiac health in this breed.

This study aims to provide standard echocardiographic measurements for Quarter Horses and evaluate the relationship between physical characteristics (body weight, age, sex) and echocardiographic measurements in this study cohort. It was hypothesized that greater body weight would be associated with larger cardiac size, while there would not be a detectable association of age and sex with echocardiographic variables.

2 | Materials and Methods

A cross-sectional study was performed evaluating sedentary Quarter Horses from a university teaching and research herd that were available for echocardiographic evaluation between May 2020 and May 2022. All available Quarter Horses were considered for inclusion in the study. Horses were housed in turnout paddocks with the ability to walk 24 h per day but did not undergo controlled exercise. All echocardiograms were performed by one of two veterinarians experienced in equine echocardiography (JMM, EWL). Horse body weight, age, and sex were documented to examine the effect of these factors on echocardiographic variables. Horses with a pre-existing history of cardiac disease and horses with moderate to severe valvular regurgitation detected during their echocardiogram were excluded from further analysis. Medical records were reviewed for a history of cardiac or pulmonary disease, and horses with a history of disease known to result in cardiac changes were excluded from subsequent analysis. All procedures were approved by an institutional animal care and use committee.

2.1 | Echocardiography

Standard unsedated echocardiographic exams were performed between May 2020 and May 2022. Images were obtained with a portable ultrasound machine (LOGIQ e, Sound, Carlsbad, CA) equipped with a 3SC-RS sector transducer with a frequency range of 1.1–4.7 MHz and simultaneous one-lead surface electrocardiogram using a standard electrode placement [1, 17, 18]. The hair was clipped bilaterally and prepared with alcohol and ultrasound gel. Routine transthoracic two-dimensional (2D), motion mode (M-mode), and color Doppler echocardiography were performed to assess cardiac structure, size, and function. The right parasternal long-axis images included a four-chamber (4C) view, a left ventricular outflow tract (LVOT) view, and a right ventricular outflow tract (RVOT) view. The left parasternal long-axis images included a view of the left atrium (L-LA), aortic root (L-AO), and pulmonic valve (L-PA). The right parasternal short-axis images included a view of the left ventricle at the level of the chordae tendineae (SAX-LV), a mitral valve (SAX-MV) view, and an aortic valve (SAX-AO) view. A minimum of one cine-loop with a minimum of three cardiac cycles per cine-loop was stored of each view for subsequent analysis. M-mode views of the left ventricle at the level of chordae tendineae, mitral valve, and aorta were stored as a minimum of three still images per view for future analysis. Color Doppler views of the mitral, aortic, tricuspid, and pulmonic valves were obtained from the right parasternal window in long-axis. An additional color Doppler view of the mitral valve was obtained from the left parasternal long-axis view.

Measurements were performed on three cardiac cycles using the leading edge-to-leading edge method, and a mean value was used for all subsequent analysis. Cycles immediately following arrhythmias, such as second-degree atrioventricular blocks, were excluded from analysis. Measurements were made on consecutive cardiac cycles. When consecutive cardiac cycles were not available or inappropriate for inclusion due to image quality or a disruption in the cardiac rhythm, an alternative cycle from the same cine-loop was included. All measurements were made by the same experienced operator (JMM) to minimize variability. Image quality was subjectively variable and dependent on both horse conformation and body condition. An instantaneous R to R interval was measured from the cycle preceding each analyzed cycle in the M-mode SAX-LV view. Left ventricular structure and size were evaluated by measuring left ventricular internal diameter (R-LVID), interventricular septum thickness (R-IVS), and left ventricular free wall thickness (R-LVFW) using the M-mode SAX-LV. Left ventricular ejection time (LVET) was measured from the M-mode SAX-AO view. Relative wall thickness (RWT_d), mean wall thickness (MWT_d), and fractional shortening (FS) were calculated using the following equations, respectively:

$$RWTd = \frac{(IVSd + LVFWd)}{LVIDd} \qquad MWTd = \frac{(IVSd + LVFWd)}{2} \qquad \%FS = \frac{LVIDd - LVIDs}{LVIDd} \times 100 \quad [7].$$

Measurements of left atrial diameter were obtained from two standard echocardiographic views: 4C view ($R\text{-}LAD_{max}$) and L-LA view ($L\text{-}LAD_{max}$). Measurements of left atrial appendage diameter ($R\text{-}LAAD_s$) were obtained using M-mode SAX-AO view. Measurements of aortic diameter were obtained from three standard echocardiographic views: LVOT view at the level

of the sinus (R-AoD$_d$), SAX-AO view (R-M-mode-AoD$_d$), and the L-AO view (L-AoD$_d$). Pulmonary artery dimensions were obtained using two standard echocardiographic views: RVOT view (R-PAD$_d$) and L-PA view (L-PAD$_d$).

Left atrial to aortic ratios were calculated as L-LAD$_{max}$/R-AoD$_d$. Left atrial to left ventricular ratios were calculated as L-LAD$_{max}$/R-LVID$_d$. Pulmonic artery to aortic ratios were calculated as R-PAD$_d$/R-AoD$_d$.

2.2 | Data Analysis

Data analysis was performed using commercially available software (GraphPad Prism 9.3.1, GraphPad Software, San Diego, CA). Continuous variables were assessed for normality using a Shapiro–Wilk test. Normally distributed data are reported as mean ± standard deviation and non-normally distributed data as median, 25th quantile, and 75th quantile. Subsequently, all length measurements were allometrically scaled to correct the influence of body size on cardiac dimensions. Measurements were normalized to a body weight of 500 kg using the following equation: $\text{diameter}_{(500\ \text{kg})} = \text{measured diameter}/\text{BWT}^{1/3} \times 500^{1/3}$ [19]. Allometrically scaled values are distinguished from the raw measurements by a (500 kg) subscript.

For each variable 95% reference ranges were calculated using a dedicated software package, as previously described (Reference Value Advisor v2.1, Toulouse, France) and employing the standard method: $m \pm t_{N-1}^{0.975}\sigma\sqrt{\frac{N+1}{N}}$, where N is the sample size, m is the sample mean, σ is the sample standard deviation, and $t_{N-1}^{0.975}$ the 0.975 quantile of a Student distribution with N-1° of freedom [3, 8]. Sample size of 40 was considered adequate for reference interval calculation [20, 21]. The distribution of each variable was assessed using raw data box plots, histograms, and normal probability plots. For all raw measurements (not corrected for body weight), untransformed data were used and outliers were accepted after visual confirmation and individual evaluation of each measurement. Untransformed data were used for reference interval calculations where symmetrical distribution of the data could be assumed. For body weight-corrected data, in cases where symmetrical distribution could not be assumed, reference intervals were calculated using Box-Cox transformed data. In all cases, outliers identified by the software were individually assessed to ensure they were suitable for inclusion and were retained in the analysis.

Multiple linear regression, using the least squares method, was selected to evaluate the relationship of multiple independent variables on a single dependent variable, while also providing information about the impact each independent variable provides individually. Unscaled body weight, age, and sex were used as independent variables and the echocardiographic measurement as the dependent variable. Mare was selected as the reference for use in regression. The two ovariectomized females were excluded from sex analysis and multiple linear regression. Simple linear regression analysis of each echocardiographic variable and the associated predictor variable (body weight or age) was subsequently performed to visualize individual correlations. For simple linear regression with age as the predictor, the allometrically scaled echocardiographic variables were used to account for the effects of body weight. Allometrically scaled echocardiographic variables were compared between mares and geldings with a Student's *t*-test or Mann–Whitney as appropriate. A p value ≤ 0.05 was considered statistically significant in all comparisons with no correction for multiple comparisons.

3 | Results

Forty-six Quarter Horses were available for inclusion at the time of the study. Four horses were excluded due to moderate to severe valvular regurgitation detected during echocardiographic evaluation, and 1 horse was excluded due to a diagnosis of asthma identified on record review. The remaining study cohort consisted of 41 Quarter Horses (25 mares, 2 ovariectomized mares, and 14 geldings), aged from 3 to 28 (median:14 years, IQR: 11–18 years), body weight 420–684 kg (mean ± SD: 541.8 ± 51.6 kg; Figure 1). All echocardiographic variables were normally distributed. Echocardiographic variables and associated reference intervals are summarized in Table 1.

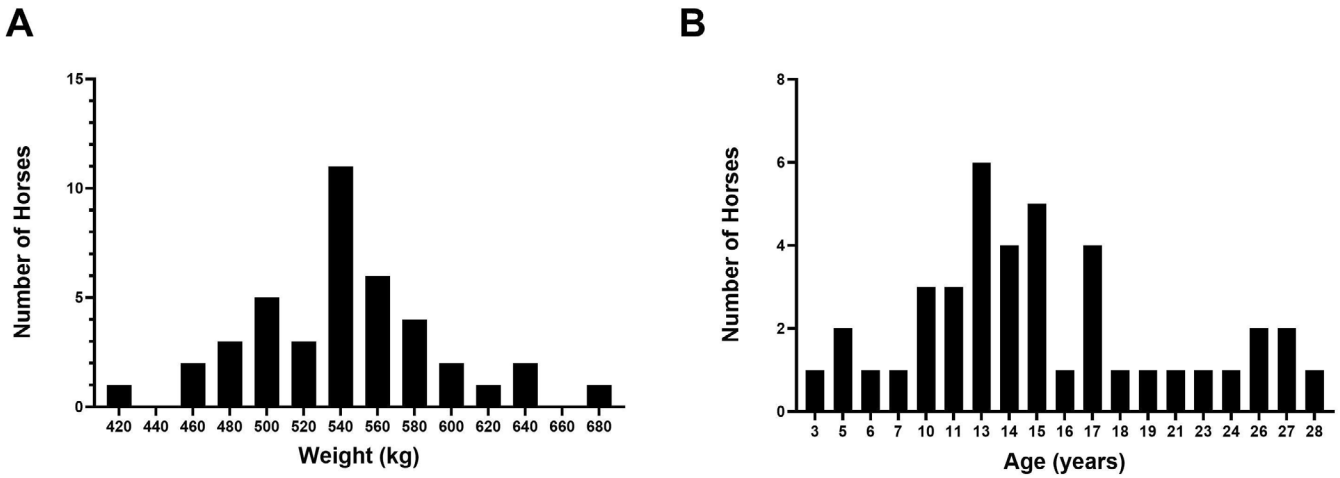

FIGURE 1 | Distribution of body weight (A) and age (B) in this study cohort of 41 Quarter Horses. Histogram A displays the number of horses by body weight binned into 20 kg groups such that the 420 kg group includes horses from 410 to 429.9 kg. Histogram B displays the number of horses in each age range by year.

Multiple linear regression performed on the unscaled measurements (Tables 2–4) showed a positive association between body weight and left ventricular free wall thickness in systole (R-LVFW$_s$), left atrial diameter (L-LAD$_{max}$ and R-LAD$_{max}$), and pulmonary artery diameter (L-PAD$_d$). Left ventricular free wall thickness in systole (R-LVFW$_s$) was 3 (95% CI 1–6) mm larger for every 100 kg in body weight ($p = 0.02$). L-LAD$_{max}$ was 8 (95% CI 2–13) mm larger for every 100 kg in body weight

TABLE 1 | Summary statistics of echocardiographic variables and associated reference intervals in Quarter Horses.

		Uncorrected measurements			Allometrically scaled (500 kg)		
Variable	***n***	**Mean ± SD**	**Lower limit of reference interval (90% CI)**	**Upper limit of reference interval (90% CI)**	**Mean ± SD**	**Lower limit of reference interval (90% CI)**	**Upper limit of reference interval (90% CI)**
M-mode echocardiography measurements							
R-IVS$_d$ (cm)	41	2.9 ± 0.3	2.3 (2.1–2.4)	3.6 (3.4–3.7)	2.9 ± 0.3	2.2 (2.1–2.3)	3.5 (3.4–3.7)
R-IVS$_s$ (cm)	41	4.4 ± 0.4	3.5 (3.3–3.7)	5.3 (5.1–5.5)	4.3 ± 0.4	3.4 (3.2–3.6)	5.2 (5.0–5.4)
R-LVID$_d$ (cm)	41	10.1 ± 0.7	8.6 (8.2–8.9)	11.6 (11.2–11.9)	9.8 ± 0.7	8.4 (8.1–8.7)	11.3 (10.9–11.6)
R-LVID$_s$ (cm)	41	5.7 ± 0.7	4.2 (3.8–4.5)	7.2 (6.9–7.5)	5.6 ± 0.7	4.1 (3.8–4.5)	7.0 (6.6–7.2)
R-LVFW$_d$ (cm)	41	2.6 ± 0.3	1.9 (1.8–2.1)	3.3 (3.1–3.5)	2.5 ± 0.3	1.9 (1.7–2.0)	3.2 (3.0–3.3)
R-LVFW$_s$ (cm)	41	4.1 ± 0.4	3.3 (3.2–3.5)	4.9 (4.7–5.0)	4.0 ± 0.3	3.1 (2.7–3.4)	4.5 (4.4–4.6)
R-LAAD$_s$(cm)	41	4.2 ± 0.9	2.3 (1.9–2.8)	6.1 (5.6–6.5)	4.1 ± 0.9	2.3 (1.9–2.7)	5.9 (5.5–6.3)
R-M-mode-AoD$_d$ (cm)	41	6.6 ± 0.5	5.5 (5.3–5.8)	7.6 (7.4–7.9)	6.4 ± 0.5	5.4 (5.2–5.6)	7.4 (7.2–7.6)
LVET (ms)	41	450 ± 40	380 (360–400)	530 (510–550)	—	—	—
RR (ms)	41	1600 ± 250	1090 (970–1210)	2110 (2000–2220)	—	—	—
Two-dimensional echocardiography measurements							
R-LAD$_{max}$ (cm)	41	9.8 ± 0.9	7.9 (7.5–8.3)	11.6 (11.2–12.0)	9.5 ± 0.8	7.8 (7.5–8.2)	11.2 (10.8–11.6)
R-AoD$_d$ (cm)	41	6.4 ± 0.5	5.3 (5.1–5.6)	7.5 (7.2–7.7)	6.2 ± 0.5	5.2 (4.9–5.4)	7.3 (7.1–7.6)
R-PAD$_d$ (cm)	40	5.1 ± 0.5	4.0 (3.8–4.3)	6.2 (5.9–6.4)	5.0 ± 0.5	4.0 (3.7–4.2)	6.0 (5.7–6.2)
L-LAD$_{max}$ (cm)	40	10.4 ± 0.8	8.9 (8.5–9.2)	12.0 (11.7–12.4)	10.2 ± 0.7	8.7 (8.4–9.1)	11.6 (11.3–12.0)
L-AoD$_d$ (cm)	40	6.4 ± 0.6	5.2 (4.9–5.5)	7.7 (7.4–8.0)	6.3 ± 0.6	5.0 (4.8–5.3)	7.6 (7.3–7.8)
L-PAD$_d$ (cm)	40	6.1 ± 0.6	4.9 (4.6–5.1)	7.3 (7.0–7.6)	5.9 ± 0.5	4.8 (4.6–5.1)	7.0 (6.8–7.3)
Calculated variables							
RWT$_d$ (cm)	41	0.6 ± 0.1	0.4 (0.4–0.4)	0.7 (0.7–0.7)	—	—	—
MWT$_d$ (cm)	41	2.8 ± 0.3	2.2 (2.1–2.3)	3.3 (3.2–3.5)	2.7 ± 0.3	2.2 (2.0–2.3)	3.2 (3.1–3.4)
FS (%)	41	43.6 ± 5.0	33.3 (30.8–35.5)	53.8 (51.7–56.1)	—	—	—
L-LAD$_{max}$/R-AoD$_d$ (cm)	40	1.6 ± 0.1	1.3 (1.3–1.4)	1.9 (1.9–2.0)	—	—	—
L-LAD$_{max}$/R-LVID$_d$ (cm)	40	1.0 ± 0.1	0.9 (0.8–0.9)	1.2 (1.2–1.2)	—	—	—
R-PAD$_d$/R-AoD$_d$ (cm)	40	0.8 ± 0.1	0.6 (0.6–0.7)	1.0 (0.9–1.0)	—	—	—

Note: For detailed explanation of echocardiographic indices see Data S1. Allometric scaling was performed with the equation length$_{(500\ kg)}$ = measured length/BWT$^{1/3}$ × 500$^{1/3}$.
Abbreviations: CI, confidence interval; *n*, number of horses; SD, standard deviation.

TABLE 2 | Results of multiple linear regression analysis performed on unscaled motion-mode echocardiographic measurements in 41 Quarter Horses evaluating the association with age, body weight (BWT), and sex (mare is the reference category).

Dependent variable	*n*	Variable	β coefficient	95% CI	*p*	R^2
M-mode echocardiography measurements						
R-IVS$_d$	39	Intercept	2.77	1.35–4.20	<0.001	0.01
		BWT (kg)	0.00	0.00–0.00	0.84	
		Age (years)	0.00	−0.02 to 0.02	0.99	
		Sex [G]	0.06	−0.22 to 0.34	0.66	
R-IVS$_s$	39	Intercept	3.11	1.38–4.83	<0.001	0.18
		BWT (kg)	0.00	0.00–0.01	0.21	
		Age (years)	0.02	0.00–0.04	0.08	
		Sex [G]	−0.30	−0.64 to 0.04	0.08	
R-LVID$_d$	39	Intercept	8.19	5.20–11.2	<0.001	0.14
		BWT (kg)	0.00	0.00–0.01	0.29	
		Age (years)	0.01	−0.03 to 0.05	0.51	
		Sex [G]	0.36	−0.24 to 0.95	0.23	
R-LVID$_s$	39	Intercept	4.38	1.48–7.28	<0.01	0.19
		BWT (kg)	0.00	0.00–0.01	0.38	
		Age (years)	−0.01	−0.05 to 0.03	0.66	
		Sex [G]	0.49	−0.09 to 1.1	0.10	
R-LVFW$_d$	39	Intercept	1.30	−0.10 to 2.7	0.07	0.11
		BWT (kg)	0.00	0.00–0.00	0.10	
		Age (years)	0.01	−0.01 to 0.03	0.30	
		Sex [G]	−0.01	−0.28 to 0.27	0.96	
R-LVFW$_s$	39	Intercept	2.18	0.71–3.7	<0.01	0.17
		BWT (kg)	**0.00**	**0.00–0.01**	**0.02**	
		Age (years)	0.01	−0.01 to 0.03	0.33	
		Sex [G]	−0.13	−0.42 to 0.16	0.36	
R-LAAD$_s$	39	Intercept	2.37	−1.60 to 6.33	0.23	0.03
		BWT (kg)	0.00	0.00–0.01	0.34	
		Age (years)	0.00	−0.05 to 0.06	0.90	
		Sex [G]	−0.31	−1.10 to 0.47	0.43	
R-M-mode-AoD$_d$	39	Intercept	4.33	2.43–6.24	<0.001	0.23
		BWT (kg)	0.00	0.00–0.01	0.06	
		Age (years)	**0.03**	**0.00–0.05**	**0.04**	
		Sex [G]	0.08	−0.30 to 0.46	0.67	

Note: For detailed explanation of echocardiographic indices see Data S1. The level of significance was defined as $p \leq 0.05$ for all null hypotheses. Values in bold are cardiac dimensions for which a significant linear relationship was identified.
Abbreviations: BWT, body weight; CI, confidence interval; G, gelding; R^2, coefficient of determination of the statistical model.

($p < 0.01$). R-LAD$_{max}$ was 9 (95% CI 3–16) mm larger for every 100 kg in body weight ($p < 0.01$). Pulmonary artery diameter (L-PAD$_d$) was 4 (95% CI 0–9) mm larger for every 100 kg in body weight ($p = 0.04$). Simple linear regression for variables associated with body weight is presented in Figure 2. There were no longer statistically significant associations detected between body weight and echocardiographic variables once they were allometrically scaled to 500 kg.

Multiple linear regression performed on the unscaled measurements (Tables 2–4) showed a positive association between age and left atrial diameter (L-LAD$_{max}$), aortic diameter (R-AoD$_d$, L-AoD$_d$, and R-M-mode-AoD$_d$), and pulmonary artery diameter (R-PAD$_d$ and L-PAD$_d$). L-LAD$_{max}$ was 0 (95% CI 0–1) mm larger for every year of age ($p = 0.02$). R-AoD$_d$ was 0 (95% CI 0–1) mm larger for every year of age ($p < 0.01$). L-AoD$_d$ was 0 (95% CI 0–1) mm larger for every year of age ($p = 0.01$). M-mode measurement of aortic diameter (R-M-mode-AoD$_d$) was 0 (95% CI 0–1) mm larger for every year of age ($p = 0.04$). Pulmonary artery diameter (R-PAD$_d$ and L-PAD$_d$) was 0 (95% CI 0–1) mm larger for every year of age ($p < 0.01$ and $p = 0.04$ respectively). Individual linear regression of allometrically scaled variables associated with age is displayed in Figure 3.

Multiple linear regression performed on the unscaled measurements (Tables 2–4) showed a statistically significant association between sex and left atrial to left ventricular ratio (L-LAD$_{max}$/R-LVID$_d$). L-LAD$_{max}$/R-LVID$_d$ was 1 (95% CI −1 to 0) mm lower in geldings compared to mares in the multivariable analysis ($p = 0.03$) and approached but did not reach statistical significance in univariable analysis ($p = 0.07$, Figure 4).

TABLE 3 | Results of multiple linear regression analysis performed on unscaled 2-dimensional echocardiographic measurements in Quarter Horses evaluating the association with age, body weight (BWT), and sex (mare is the reference category).

Dependent variable	*n*	Variable	β coefficient	95% CI	*p*	R^2
Two-dimensional echocardiography measurements						
R-LAD$_{max}$	39	Intercept	4.22	0.93–7.50	0.01	0.27
		BWT (kg)	**0.01**	**0.00–0.02**	**< 0.01**	
		Age (years)	0.04	0.01–0.08	0.10	
		Sex [G]	−0.54	−1.19 to 0.11	0.10	
R-AoD$_d$	39	Intercept	4.85	2.90–6.80	< 0.001	0.24
		BWT (kg)	0.00	0.00–0.01	0.35	
		Age (years)	**0.04**	**0.01–0.07**	**< 0.01**	
		Sex [G]	0.12	−0.26 to 0.51	0.53	
R-PAD$_d$	38	Intercept	2.94	0.93–4.95	< 0.01	0.25
		BWT (kg)	0.00	0.00–0.01	0.12	
		Age (years)	**0.04**	**0.01–0.06**	**0.01**	
		Sex [G]	0.11	−0.29 to 0.50	0.59	
L-LAD$_{max}$	38	Intercept	5.70	2.80–8.59	< 0.001	0.29
		BWT (kg)	**0.01**	**0.00–0.01**	**< 0.01**	
		Age (years)	**0.05**	**0.01–0.09**	**0.02**	
		Sex [G]	−0.31	−0.88 to 0.26	0.27	
L-AoD$_d$	38	Intercept	5.18	2.84–7.51	< 0.001	0.18
		BWT (kg)	0.00	0.00–0.01	0.57	
		Age (years)	**0.04**	**0.01–0.08**	**0.01**	
		Sex [G]	0.08	−0.38 to 0.54	0.72	
L-PAD$_d$	38	Intercept	3.21	0.94–5.49	< 0.01	0.25
		BWT (kg)	**0.00**	**0.00–0.01**	**0.04**	
		Age (years)	**0.03**	**0.00–0.07**	**0.04**	
		Sex [G]	0.10	−0.35 to 0.55	0.65	

Note: For detailed explanation of echocardiographic indices see Data S1. The level of significance was defined as $P \leq 0.05$ for all null hypotheses. Values in bold are cardiac dimensions for which a significant linear relationship was identified.
Abbreviations: BWT, body weight; CI, confidence interval; G, gelding; R^2, coefficient of determination of the statistical model.

4 | Discussion

This study established normal echocardiographic variables for a cohort of sedentary Quarter Horses and identified relationships between specific echocardiographic variables, body weight, age, and sex. While extensive work has been done to establish reference intervals for horses, the Quarter Horse is underrepresented [3, 8, 9, 17, 22–24]. Limited studies evaluating Quarter Horse echocardiography include reports in athletic Quarter Horses from São Paulo, Brazil, intercollegiate reiners at Texas A&M, and single case reports or case series documenting specific abnormalities identified in Quarter Horses [11–16]. Exercise testing and electrophysiology studies have been performed in barrel horses and reining horses, which are typically Quarter Horse or Quarter Horse breed crosses, but extensive echocardiographic studies in these breeds are lacking [25, 26]. Quarter Horses are typically heavily muscled, but shorter and stockier in size than Thoroughbreds, Standardbreds, or Warmbloods, resulting in a more challenging conformation for echocardiography. Echocardiographic variables reported here are comparable to those reported in athletic Quarter Horses [11, 12], but smaller than reported for Thoroughbreds, Standardbreds, or Warmbloods, even when allometrically scaled to account for body weight [3, 7–9]. Athletic conditioning has varying effects on heart structure and function across animal species, suggesting the effect of athletic conditioning on cardiac structure might not influence these echocardiographic variables as profoundly as in other species [27–29] or horses training in other disciplines [30–33]. The Quarter Horses included in this study included a diverse group of horses with a broad age and body weight distribution representative of the diversity within the breed. Overall, these findings provide a reference to facilitate cardiac evaluation in a wide range of Quarter Horses. This will allow for more robust evaluation of the less athletic population of Quarter Horses that make up much of the recreational riding horses in the United States.

Allometric scaling of echocardiographic variables can correct for differences in body weight within a species and facilitate evaluation of cardiac chamber measurements without the influence of horse size [19, 34]. Allometric scaling of cardiac measurements is used across different breeds of adult dogs and has been successfully established in several breeds of horses, including Warmbloods, Thoroughbreds, Arabians, and Standardbreds [24, 35]. While this study did not have enough young horses to evaluate breed-specific scaling exponents for Quarter Horses, the relationship between body weight and

TABLE 4 | Results of multiple linear regression analysis performed on unscaled calculated echocardiographic measurements in Quarter Horses evaluating the association with age, body weight (BWT), and sex (mare is the reference category).

Dependent variable	*n*	Variable	β coefficient	95% CI	*p*	R^2
Calculated variables						
RWT_d	39	Intercept	0.53	0.23–0.82	< 0.01	0.01
		BWT (kg)	0.00	0.00–0.00	0.87	
		Age (years)	0.00	0.00–0.00	0.90	
		Sex [G]	−0.01	−0.07 to 0.05	0.69	
MWT_d	39	Intercept	2.02	0.83–3.22	< 0.01	0.06
		BWT (kg)	0.00	0.00–0.00	0.27	
		Age (years)	0.00	−0.01 to 0.02	0.54	
		Sex [G]	0.03	−0.21 to 0.26	0.82	
FS	39	Intercept	46.5	26.6–66.4	< 0.001	0.16
		BWT (kg)	−0.01	−0.04 to 0.03	0.67	
		Age (years)	0.16	−0.10 to 0.42	0.23	
		Sex [G]	−2.74	−6.67 to 1.19	0.17	
$L\text{-}LAD_{max}/R\text{-}AoD_d$	38	Intercept	1.35	0.76–1.94	< 0.001	0.07
		BWT (kg)	0.00	0.00–0.00	0.22	
		Age (years)	0.00	−0.01 to 0.00	0.39	
		Sex [G]	−0.07	−0.19 to 0.05	0.23	
$L\text{-}LAD_{max}/R\text{-}LVID_d$	38	Intercept	0.80	0.51–1.09	< 0.001	0.19
		BWT (kg)	0.00	0.00–0.00	0.15	
		Age (years)	0.00	0.00–0.01	0.16	
		Sex [G]	**−0.06**	**−0.12 to −0.01**	**0.03**	
$R\text{-}PAD_d/R\text{-}AoD_d$	38	Intercept	0.65	0.28–1.01	< 0.01	0.03
		BWT (kg)	0.00	0.00–0.00	0.43	
		Age (years)	0.00	0.00–0.01	0.70	
		Sex [G]	0.00	−0.08 to 0.07	0.93	

Note: For detailed explanation of echocardiographic indices see Data S1. The level of significance was defined as $P \leq 0.05$ for all null hypotheses. Values in bold are cardiac dimensions for which a significant linear relationship was identified.
Abbreviations: BWT, body weight; CI, confidence interval; G, gelding; R^2, coefficient of determination of the statistical model.

specific variables validates the utility of scaling in sedentary Quarter Horses. Allometrically scaled variables offer a simplified method of interpreting echocardiographic measurements, making echocardiographic interpretation more accessible to practitioners that perform a limited number of echocardiograms. Despite the functionality of allometric scaling in this study, caution during interpretation should still be used since there is a large variation within the Quarter Horse breed and breed-specific scaling exponents have not been established.

Age was associated with larger great vessels (aorta and pulmonary artery) and left atria in this study. The effects of age on great vessels are consistent with that in the horse in general as well as Coldblooded Trotters specifically [2, 22]. The increase in left atrial diameter with age occurs in Coldblooded Trotters [22]. However, the Coldblooded Trotters are generally younger and fitter, making growth and training more of a factor than in this sedentary cohort of Quarter Horses. Based on the age range (3–28 years) and sedentary nature of our study cohort, it is more likely these are true aging changes rather than effects of training or growth. However, growth effects cannot be ruled out in the younger animals. Age stratification of normal variables in future large-scale studies might be helpful in mitigating the effects of age on echocardiographic interpretation.

The effects of sex were evaluated in the study but limited to mares and geldings. Left ventricular size is larger in males compared to female horses, consistent with the smaller left atrial to left ventricular ratio in geldings in this study [12, 22]. This difference in left ventricular size was not identified in endurance horses [33]. However, males in the endurance horse study had smaller left atrial dimensions, which is also consistent with the smaller left atrial to left ventricular ratio in this study [34]. While sex had an influence on $L\text{-}LAD_{max}/R\text{-}LVID_d$ in this study, the effects of sex are small relative to the effects of age and body weight on echocardiographic variables.

This study contains several limitations. The diversity of the Quarter Horses in this study added to the broad applicability but might limit the application of these reference values to specific subsets of Quarter Horses bred for individual disciplines. Sex was evaluated in this study; however, this cohort lacked stallions, and thus additional differences between sexes could exist and should be investigated. Additionally, this study relied on body weight as a measure of body type. Alternative measures such as height, heart-girth, or body condition score might have a stronger association with echocardiographic measurements but were not performed in this study [3, 35]. These measurements are less routinely collected in horses in a hospital setting and

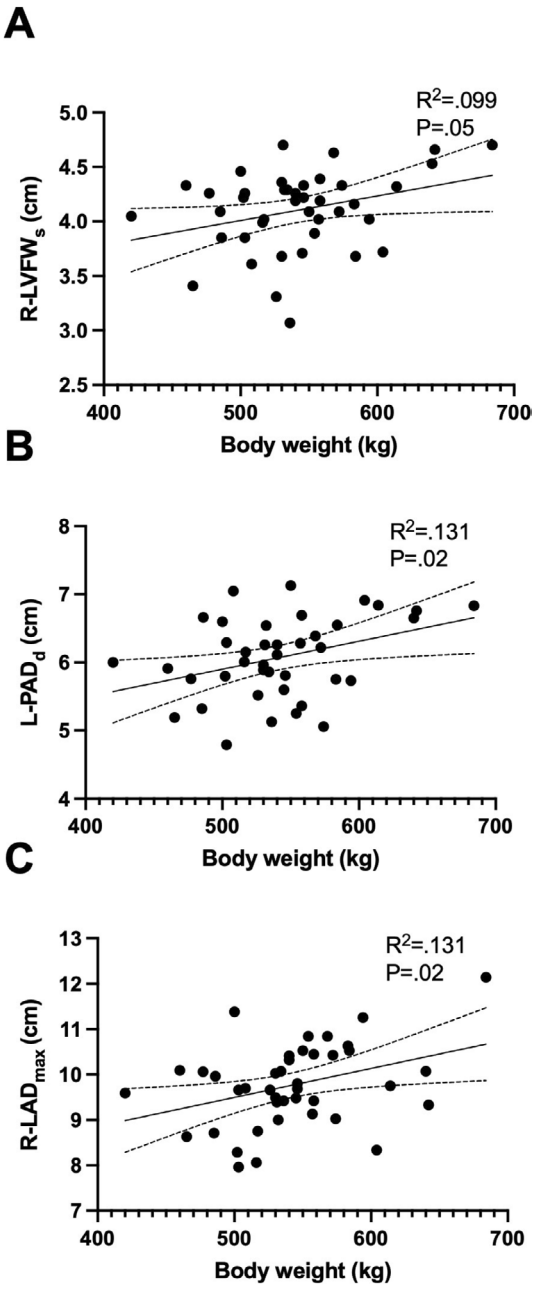

FIGURE 2 | Body weight associated echocardiographic variables for Quarter Horses. Echocardiographic variables versus body weight (kg) with a linear regression line and the 95% confidence interval graphed by individual for left ventricular free wall thickness at end-systole measured from M-mode SAX-LV view (R-LVFW$_s$, A, $n = 41$), pulmonary artery diameter at end-diastole measured from L-PA view (L-PAD$_d$, B, $n = 40$), and left atrial diameter at end-systole measured from 4C view (R-LAD$_{max}$, C, $n = 41$).

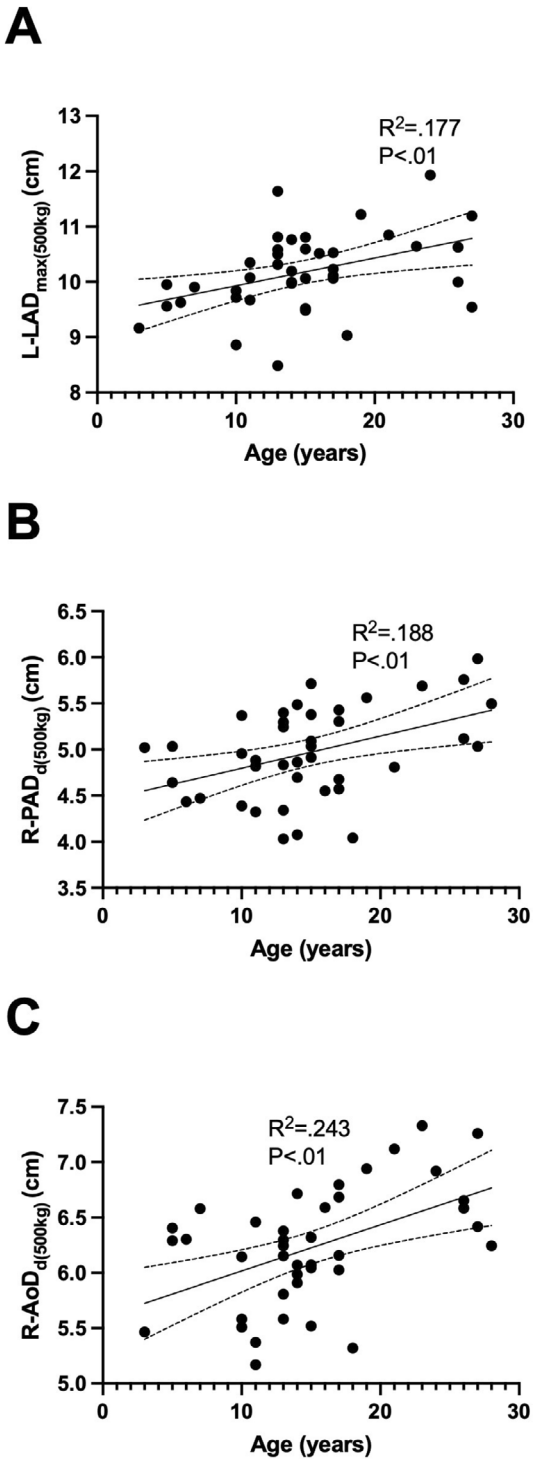

FIGURE 3 | Age associated echocardiographic variables for Quarter Horses. Allometrically scaled echocardiographic variables versus age (years) with a linear regression line and the 95% confidence interval is graphed by individual for left atrial diameter at end-systole measured from L-LA view (L-LAD$_{max(500\ kg)}$, A, $n = 40$), pulmonary artery diameter at end-diastole measured from RVOT view (R-PAD$_{d(500\ kg)}$, B, $n = 40$), and aortic diameter measured at end-diastole from LVOT view (R-AoD$_{d\ (500\ kg)}$, C, $n = 41$).

therefore are less practical than body weight. However, height and heart-girth have advantages over body weight because they are less likely to vary over the lifetime of the horse.

Standardizing echocardiographic variables for healthy non-athletic Quarter Horses provides improved reference values to facilitate the assessment of cardiac health. The results of this study support correcting for body weight in echocardiographic measurements for Quarter Horses. Older age was associated with larger measurements of the great vessels (aorta and pulmonary artery) and left atrial diameter in this study. Geldings had a smaller left atrial to left ventricular ratio compared to mares. Ultimately, understanding the association of body weight, age,

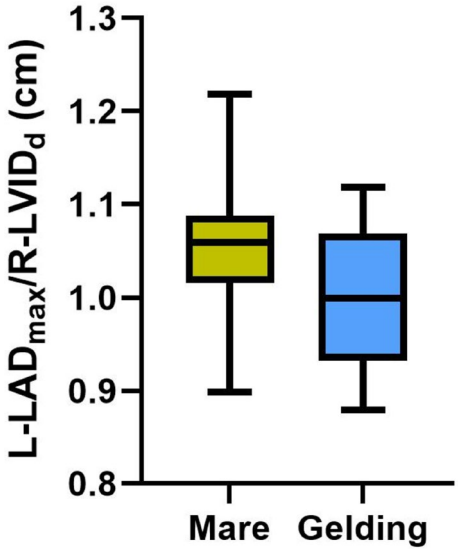

FIGURE 4 | Sex associated echocardiographic variables for Quarter Horses. Box-and-whisker plot showing median and interquartile range of left atrial to left ventricular ratio (L-LAD$_{max}$/R-LVID$_d$, $p = 0.03$) for mares and geldings. L-LAD$_{max}$/R-LVID$_d$ showed a statistically significant association with sex in the multivariable model but did not reach significance in univariable analysis displayed here ($p = 0.07$). $N = 38$ (24 mares and 14 geldings).

and sex with cardiac dimensions will provide insight into the physical characteristics that affect echocardiographic assessment and aid veterinarians assessing cardiac disease in the Quarter Horse.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by University of California Davis Institutional Animal Care and Use Committee, protocol # 22800. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** jvim70237-sup-0001-Supinfo1.docx. **Data S2:** jvim70237-sup-0002-DataS2.xlsx.